Belief Verbs Increase Perceptions of AI Animacy during Semantic Integration, Goal Verbs Do Not

Introduction. Placing non-animates in the agentive subject position of active transitive clauses, a structure known as grammatical metaphor (GM),[1] creates temporary animacy violations that must be resolved during online processing.[2] GM also influences later controlled judgments by increasing the perceived agency and responsibility of inanimate subjects[3,4] and by reducing negative agency bias (NAB), the tendency to attribute negative outcomes to human agents.[5] These findings suggest that GM may elicit anthropomorphic interpretations during situation model construction. However, it is not known if it causes the parser to enrich representations of inanimate subjects with *specific* mental-state features during semantic integration.

The current study tests whether belief vs. goal verbs used in GM sentences about artificial intelligence (AI) trigger elaborative inferences about the other mental-state property during online processing and in later judgments. Intentional stance theory[6] predicts that any anthropomorphic attribution should license broader inferences about core agent properties, whereas mind perception theory[7] predicts that belief and goal attributions can be made independently. AI was chosen as the inanimate subject because it is grammatically non-animate but has some pseudo-agentive features, allowing us to test whether GM prompts the parser to enrich non-animate subjects with specific mental-state features during semantic integration.

Stimuli. The experiment followed a 3 [Framing Condition: No-GM, Belief-GM, Goal-GM] x 3 [Explanation Condition: Inanimate, Belief Verb, Goal Verb] x 2 [Outcome: Positive, Negative] design. We constructed 24 short stories about AI, each with three versions (belief, goal, and no-GM framing). Framing was manipulated by varying whether the AI appeared as a causal adjunct (No-GM) or as the grammatical subject of either a belief or a goal verb (Belief-GM, Goal-GM). Belief and goal verbs were selected from a norming study in which 250 Prolific participants rated 54 animate verbs on belief- and goal-related features. 24 verbs with unambiguous ratings were used (Figure 1). Each story was followed by an explanation sentence of the form “It all happened because [the AI] had [critical verb] [spillover region]”. Across conditions, explanation sentences differed only in the critical verb. Items were matched on length, syntactic structures, and content words. Participants never saw the same critical verb more than once.

Results. 61 undergraduate participants completed the eye-tracking experiment using Eyelink 1000[8]. Framing was manipulated between participants, while explanation type was within participants. We collected agreement ratings and late eye-tracking measures—fixation count (FC), dwell time (DT), regression path duration—on the critical verb. Eye data was residualized with lexical variables (frequency, concreteness, length) prior to analysis.

In agreement ratings (Figure 2), belief framing increased agreement with all explanation types, whereas goal framing increased agreement *only* with goal explanations ($t = 3.26$, $p = .001$). Belief explanations (but not goal explanations) eliminated NAB ($t = 12.46$, $p < .001$), a pattern replicated in multiple eye-tracking measures (Figure 3): for negative outcomes, belief explanations showed reduced FC on both the critical ($t = -3.13$, $p = .002$) and spillover ($t = -3.44$, $p < .001$) regions, as well as reduced DT ($t = -3.33$, $p < .001$). DT results further indicated that participants with stronger preexisting anthropomorphic beliefs about AI were particularly sensitive to these effects ($t = -2.56$, $p = .010$).

Conclusions. Results show that linguistic framing influences readers’ reasoning about AI. Belief explanation produced an anthropomorphic reduction in NAB in both controlled judgments and online sentence-processing measures. Belief framing also increased agreement with both belief and goal explanations, whereas goal framing increased agreement only with goal explanations. The intentional stance predicted that both belief and goal verbs would entail full animacy; mind perception theory predicted that inferences of beliefs and goals would be orthogonal. Our results indicate the existence of an implicational hierarchy not predicted by either theory, in which belief verbs promote broader animacy-related enrichment during semantic integration and in later judgments, while goal verbs do not.

References

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Appendix

Fig. 1

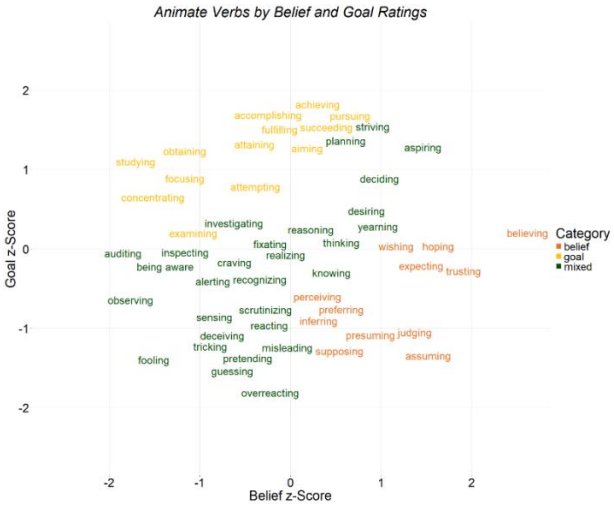

Fig. 2

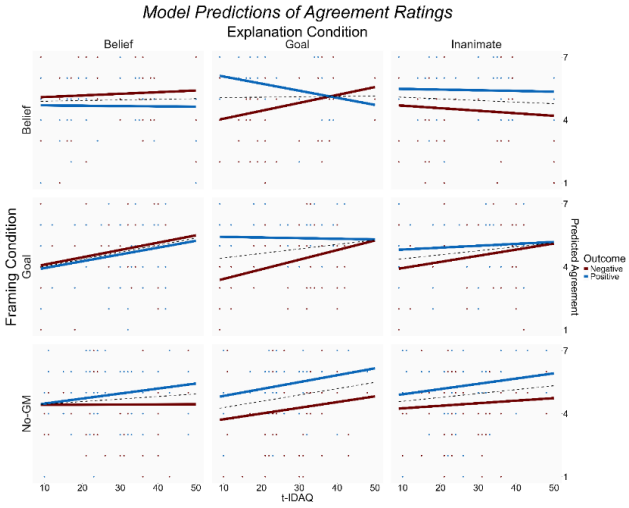

Fig. 3

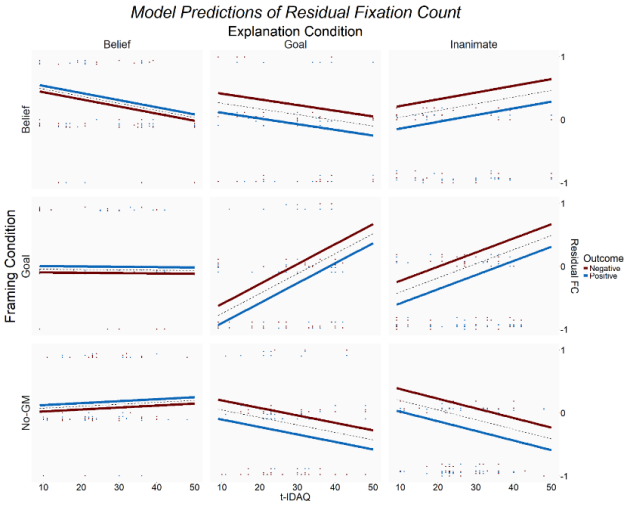

Fig. 1 Animate Verbs by mean belief (x axis) and goal (y axis) ratings. Color shows which verbs were selected as critical verbs for the eye-tracking study.

Fig. 2 Model predictions of agreement (1 = totally disagree, 7 = totally agree) by framing condition, explanation condition, outcome, and t-IDAQ[9] (a measure of technology anthropomorphism, 9 = low, 48 = high). Participants agreed more with explanations for positive vs negative stories, *except* in the belief explanation condition (and in the goal explanation condition, for high t-IDAQ participants), showing the belief explanations reduced NAB.

Fig. 3 Model predictions of residual FC by framing condition, explanation condition, outcome, and t-IDAQ. Higher residual FC indicates more fixations than predicted by lexical features (i.e., more difficult processing). It was easier for readers to process belief verbs for negative stories, but easier to process goal/inanimate verbs for positive stories, showing that belief explanations reduced NAB.